Adaptive and Learning-Guided Extensions of Semi-Sparsity for Image Smoothing

Anuj Saha
Indian Institute of Information Technology Vadodara
202351010@iiitvadodara.ac.in

Divyanshu Ghosh
Indian Institute of Information Technology Vadodara
202351036@iiitvadodara.ac.in

Gadige Nikhil
Indian Institute of Information Technology Vadodara
202351037@iiitvadodara.ac.in

***Abstract*—Edge-preserving image smoothing is a fundamental problem in image processing with various applications in denoising, abstraction, and preprocessing of images for higher-level vision tasks. Semi-sparsity methods with L_0 constraints on second-order image gradients have shown impressive potential in preserving structural edges and reducing noise. However, the use of a fixed sparsity threshold limits the effectiveness of these approaches on images with non-uniform content. In this work, we explore three extensions to the semi-sparsity framework: (i) adaptive gradient-order selection via spatially varying thresholds, (ii) learning-guided semi-sparsity (LGSS) incorporating edge-based priors, and (iii) multi-scale decomposition. The experimental results, carried out on standard benchmarks with Gaussian noise, show that the adaptive extension produces the greatest improvements, reaching 4.45 dB on the Lena image and 1.58 dB on the Cameraman image. The proposed LGSS framework uses a hybrid prior, which, despite outperforming all other methods, cannot beat the performance of the adaptive extension, given the redundancy with gradient-based information and sensitivity to edge map quality. We will also demonstrate that multi-scale extensions cannot be made compatible with the global optimization structure of the semi-sparsity formulation. These results highlight the importance of spatially adaptive regularization and provide insights into the integration of external priors within variational smoothing frameworks.**

***Index Terms*—Image smoothing, edge-preserving filtering, L_0 gradient minimization, semi-sparsity, adaptive regularization, learning-guided optimization, image denoising.**

I. Introduction

IMAGE smoothing is a basic building block in low-level computer vision and is an essential preprocessing step for various tasks, including image segmentation, object detection, feature extraction, and computational photography. The main problem is how to selectively reduce small variations due to noise or texture, while preserving large variations corresponding to semantically meaningful image structures like object boundaries and large-scale intensity changes. These two goals have been at the heart of edge-preserving image smoothing filters for several decades. The conventional methods for edge preservation can be generally divided into two categories: local filtering approaches and global optimization methods. Local filtering approaches, such as bilateral filtering [1] and guided filtering [2], are efficient because of their local nature, but they suffer from gradient reversal and parameter sensitivity issues. Global optimization methods, especially those employing total variation (TV) regularization [4], are often plagued by staircases resulting from the L_1 norm on gradients.

The introduction of L_0 gradient minimization [9] provided a paradigm shift that directly controlled the number of non-zero gradients in the smoothed image. The discrete optimization approach was able to aggressively smooth uniform regions while maintaining a very strong edge response to salient edges. The problem with the first-order approach is that it treats all gradients uniformly, resulting in poor performance for images containing structures at multiple scales.

The semi-sparsity framework [10] is an extension of this foundation, where L_0 sparsity constraints are imposed on *second-order* gradients, capturing curvature information. The higher-order regularization is useful to achieve piecewise linear approximations of the image structure, hence smoothing flat regions and linear gradients, and preserving transitions. The model is the sum of a quadratic first-order term and an L_0 second-order term, and the relative importance is determined by tuning the weights.

Despite these advances, the original semi-sparsity formulation exhibits several limitations when applied to diverse natural images:

1) **Spatially uniform thresholding**: The global threshold β applied to second-order gradients does not account for local image statistics, leading to over-smoothing in textured regions and under-smoothing in homogeneous areas.
2) **Parameter brittleness**: Performance is highly sensitive to the choice of β and the regularization weight λ, with optimal values varying significantly across different image types.
3) **Texture handling**: Images with significant texture content (e.g., Barbara) pose challenges as texture patterns are neither pure noise nor semantically significant edges.
4) **Scale limitations**: The single-scale formulation processes all image structures at a uniform resolution, potentially missing multi-scale hierarchical relationships.

This work addresses these limitations through three targeted extensions:

1) **Adaptive gradient-order selection**: We introduce a spatially varying threshold $\beta(x, y)$ modulated by local first-order gradient magnitude. This enables automatic adjustment of smoothing strength based on local edge probability, reducing parameter sensitivity.
2) **Learning-guided semi-sparsity (LGSS)**: We incorporate external edge probability maps, derived from Canny

edge detection followed by Gaussian smoothing, into the optimization framework. The edge map $E(x, y)$ is inverted to produce a weight map $W(x, y) = 1 - E(x, y)$, which modulates the local sparsity threshold as threshold$(x, y) = \frac{\beta \cdot W(x,y)}{\lambda}$.

3) **Multi-scale decomposition**: We explore a coarse-to-fine strategy where the image is first smoothed at reduced resolution, then upsampled to initialize fine-scale optimization.

Our experimental evaluation on four standard test images (Lena, Cameraman, Barbara, Strip) corrupted by additive Gaussian noise reveals that:

- The adaptive extension provides consistent improvements across diverse image types, with PSNR gains ranging from 0.02 dB (Strip) to 4.45 dB (Lena).
- LGSS incorporates externally derived edge priors but exhibits sensitivity to edge detection quality, achieving competitive results on Lena (28.88 dB) and Cameraman (27.44 dB) but degrading performance on Strip.
- The multi-scale approach fundamentally conflicts with the global FFT-based optimization structure, resulting in severe performance degradation and demonstrating that not all extensions are compatible with the underlying algorithmic framework.

Contributions. The main contributions of this work are:

- We propose an adaptive semi-sparsity approach with spatially varying thresholds, which leads to a significant improvement in image denoising.
- We introduce a learning-guided semi-sparsity approach, referred to as LGSS, where external edge information is incorporated.
- We perform an extensive analysis of various extensions, including a negative result on the incompatibility of multi-scale decomposition with semi-sparsity.
- We discuss some of the drawbacks of second-order sparsity for texture images and identify avenues for improvement.

The remainder of this paper is organized as follows. Section II reviews related work in edge-preserving filtering and sparse regularization. Section III presents the mathematical formulation of the semi-sparsity framework. Section IV details our proposed extensions and their theoretical motivations. Section V describes the experimental setup and evaluation methodology. Section VI presents comprehensive results and analysis. Section VII discusses practical applications. Section VIII concludes with insights and future directions.

II. Related Work

A. Edge-Preserving Filtering

Edge-preserving smoothing has been studied from both local and global perspectives, and many methods have been developed in this direction. The bilateral filter [1] was one of the first to use range kernels, which are adaptive to local intensity changes, and has been successful in edge-aware averaging. However, bilateral filtering is a local operation, and when edges are present, gradient reversal might occur if they are too sharp. Guided filtering [2] has also been successful in terms of achieving linear time complexity, but similar problems exist when handling edges.

Anisotropic diffusion [3] has also been successful in edge-preserving smoothing, and they use a partial differential equation to solve this problem, with diffusion depending on local gradient magnitudes. However, solving a partial differential equation involves choosing a proper time step, and instabilities may occur near edges.

B. Total Variation and L_1 Regularization

Total variation (TV) regularization, proposed by Rudin, Osher, and Fatemi in [4], aims to minimize the L_1 norm of the gradient of an image. The convex nature of TV regularization makes global optimums feasible with computationally efficient algorithms like split Bregman [5] and primal-dual algorithms [6].

TV regularization has been shown to have staircase effects, where smooth gradients in an image are replaced with piecewise constant values separated by step functions. Higher-order TV [7] and weighted TV [8] regularizers have been proposed to overcome staircase effect of TV with varying degrees of success.

C. L_0 Gradient Minimization

The L_0 gradient minimization framework [9] directly minimizes the number of non-zero gradients through a discrete optimization formulation:

$$\min_S \|S - I\|^2 + \lambda \|\nabla S\|_0 \tag{1}$$

where $\|\nabla S\|_0$ counts the number of pixels with non-zero gradients. This non-convex optimization problem is solved via half-quadratic splitting (HQS), which alternates between solving a quadratic problem with FFT and performing hard thresholding in the gradient domain.

The L_0 penalty provides superior edge preservation compared to L_1 by avoiding the bias toward zero introduced by the absolute value function. However, the first-order formulation treats all gradients uniformly, regardless of whether they correspond to texture, noise, or semantic edges.

D. Semi-Sparsity for Smoothing Filters

The semi-sparsity framework [10] extends L_0 gradient minimization by incorporating second-order derivatives. The energy functional combines a quadratic first-order term with an L_0 second-order term:

$$E(S) = \|S - I\|^2 + \alpha \|\nabla S\|^2 + \lambda \|\nabla^2 S\|_0 \tag{2}$$

This formulation promotes piecewise-linear approximations, smoothing both constant and linear gradients while preserving curvature discontinuities. The quadratic first-order term $\alpha \|\nabla S\|^2$ provides baseline smoothness, while the L_0 second-order term $\lambda \|\nabla^2 S\|_0$ enforces sparsity in the curvature domain.

The optimization is done through an HQS approach, utilizing FFT for the inversion of the quadratic subproblem. The

continuation approach is used to smoothly vary the value of λ, starting with a nearly quadratic problem and then sharpening the sparsity constraint to prevent poor local minima.

Although this approach is successful, the previous formulation uses a uniform threshold over the entire image, which is restrictive for adaptability to varied content.

E. Adaptive and Learning-Based Regularization

Recent advances in image processing have focused on adaptive regularization, where the weights are image-dependent and vary over the spatial domain [11]. The application of learning-based methods [12] shows the potential improvement in optimization performance by using external image priors, such as edges and segmentations.

Our contributions are to adapt the semi-sparsity concept and combine adaptive thresholding and the guidance of learning-based methods, inheriting the structural merits of second-order regularization.

III. Problem Formulation

A. Notation and Preliminaries

Let $I \in \mathbb{R}^{N\times M}$ denote the observed noisy image, and $S \in \mathbb{R}^{N\times M}$ represent the smoothed output. We consider the additive Gaussian noise model:

$$I(x,y) = I_0(x,y) + n(x,y) \tag{3}$$

where I_0 is the clean ground truth and $n \sim \mathcal{N}(0,\sigma^2)$ is white Gaussian noise.

The first-order gradients are defined using central differences:

$$\nabla_x S = S * D_x, \quad D_x = \frac{1}{2}[1,-1] \tag{4}$$

$$\nabla_y S = S * D_y, \quad D_y = \frac{1}{2}[1,-1]^T \tag{5}$$

The second-order gradients capture curvature along axis-aligned and diagonal directions:

$$\nabla_{xx} S = S * F_{xx}, \quad F_{xx} = \frac{1}{4}[1,-2,1] \tag{6}$$

$$\nabla_{yy} S = S * F_{yy}, \quad F_{yy} = \frac{1}{4}[1,-2,1]^T \tag{7}$$

$$\nabla_{uu} S = S * F_{uu}, \quad F_{uu} = \frac{1}{4}\begin{bmatrix}1 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & 1\end{bmatrix} \tag{8}$$

$$\nabla_{vv} S = S * F_{vv}, \quad F_{vv} = \frac{1}{4}\begin{bmatrix}0 & 0 & 1\\ 0 & -2 & 0\\ 1 & 0 & 0\end{bmatrix} \tag{9}$$

The L_0 norm counts non-zero elements:

$$\|\nabla^2 S\|_0 = |\{(x,y) : \nabla^2_{xx} + \nabla^2_{yy} + \nabla^2_{uu} + \nabla^2_{vv} > 0\}| \tag{10}$$

B. Energy Functional

The semi-sparsity energy functional combines three terms:

$$E(S) = \underbrace{\|S-I\|_F^2}_{\text{Data fidelity}} + \underbrace{\alpha\|\nabla S\|_F^2}_{\text{First-order smoothness}} + \underbrace{\lambda\|\nabla^2 S\|_0}_{\text{Second-order sparsity}} \tag{11}$$

The **data fidelity** term $\|S-I\|_F^2$ penalizes deviation from the observed image. The **first-order smoothness** term $\alpha\|\nabla S\|_F^2$ is a quadratic penalty that promotes overall smoothness without enforcing sparsity. The **second-order sparsity** term $\lambda\|\nabla^2 S\|_0$ enforces a discrete constraint on the number of non-zero second-order gradients.

The parameter α controls the baseline smoothness strength, typically set to small values (e.g., $\alpha = 0.1$). The parameter λ controls the sparsity penalty and is gradually increased during optimization. The threshold β determines which second-order gradients are set to zero during the HQS iterations.

C. Half-Quadratic Splitting (HQS)

The non-convex L_0 term in (11) is handled via HQS, which introduces auxiliary variables $g_{xx}, g_{yy}, g_{uu}, g_{vv}$ to approximate the second-order gradients. The optimization alternates between:

Step 1: S-subproblem (with fixed g). Solve the quadratic problem:

$$\begin{aligned}\min_S \|S-I\|_F^2 + \alpha\|\nabla S\|_F^2 \\ + \lambda\big(\|g_{xx} - \nabla_{xx}S\|_F^2 + \|g_{yy} - \nabla_{yy}S\|_F^2 \\ + \|g_{uu} - \nabla_{uu}S\|_F^2 + \|g_{vv} - \nabla_{vv}S\|_F^2\big)\end{aligned} \tag{12}$$

Taking the derivative and setting to zero yields a linear system that can be solved efficiently in the Fourier domain:

$$S = \mathcal{F}^{-1}\left[\frac{\mathcal{F}[I] + \alpha\mathcal{F}[\text{div}(\nabla S)] + \lambda\mathcal{F}[\text{div}^2(g)]}{1 + \alpha\text{OTF}_\nabla + \lambda\text{OTF}_{\nabla^2}}\right] \tag{13}$$

where OTF denotes the optical transfer function (Fourier transform of the difference kernels), and div denotes the divergence operator (adjoint of gradient).

Step 2: g-subproblem (with fixed S). Apply hard thresholding:

$$g(x,y) = \begin{cases}\nabla^2 S(x,y) & \text{if } \|\nabla^2 S(x,y)\|^2 \geq \frac{\beta}{\lambda}\\ 0 & \text{otherwise}\end{cases} \tag{14}$$

The threshold β/λ decreases as λ increases during the continuation scheme, progressively enforcing sparsity.

Continuation scheme. Initialize $\lambda = \lambda_0$. After each iteration, update $\lambda \leftarrow \kappa\lambda$ with $\kappa > 1$ (typically $\kappa = 1.2$) and $\alpha \leftarrow \tau\alpha$ with $\tau < 1$ (typically $\tau = 0.95$). Terminate when $\lambda \geq \lambda_{\max}$ or the maximum iteration count is reached.

The optimization problem is addressed through the use of the half-quadratic splitting (HQS) approach, which transforms the non-convex optimization problem into a sequence of tractable sub-problems. The use of auxiliary variables for the gradient terms allows for an alternating approach to sparsity through thresholding and a quadratic reconstruction problem in the Fourier domain. The problem is efficiently addressed through the use of fast Fourier transforms (FFT), and the

sparsity constraint is enforced through continuation on the regularization parameter λ.

IV. Methodology

A. Adaptive Gradient-Order Selection

The original formulation of semi-sparsity uses a global threshold β that is uniform over the entire image. The uniformity of this approach, however, might be suboptimal, since there are some image regions where strong edges should be preserved with higher thresholds, and some where uniform smoothing should be allowed with lower thresholds.

We propose an adaptive threshold $\beta(x, y)$ that varies spatially based on local first-order gradient magnitude:

$$\beta(x, y) = \beta_0 \cdot w(x, y)^\eta \tag{15}$$

where $w(x, y)$ is a normalized weight map derived from the gradient magnitude:

$$w(x, y) = \exp\left(-\gamma\sqrt{\nabla_x S^2 + \nabla_y S^2}\right) \tag{16}$$

The parameter γ controls the sensitivity to gradient magnitude, and η adjusts the dynamic range of the adaptive weights. In our implementation, we set $\gamma = 10$ and $\eta = 2$ based on empirical tuning.

Rationale: At edge locations, where $\|\nabla S\|$ is large, $w(x, y) \approx 0$, resulting in $\beta(x, y) \approx 0$. This effectively lowers the threshold $\beta(x, y)/\lambda$, making it *easier* for second-order gradients to survive the sparsity constraint, thus preserving edge sharpness. Conversely, in smooth regions where $\|\nabla S\| \approx 0$, $w(x, y) \approx 1$, yielding $\beta(x, y) \approx \beta_0$, which enforces stronger sparsity and promotes smoothing.

The adaptive mechanism is integrated into the HQS framework by replacing the global threshold in the g-subproblem:

$$g(x, y) = \begin{cases} \nabla^2 S(x, y) & \text{if } \|\nabla^2 S(x, y)\|^2 \geq \frac{\beta(x,y)}{\lambda} \\ 0 & \text{otherwise} \end{cases} \tag{17}$$

Limitations: The adaptive approach introduces two additional hyperparameters (γ, η) that require tuning. The reliance on first-order gradients from the current estimate S can be noisy in early iterations. Furthermore, the exponential weighting may be overly aggressive in suppressing thresholds near edges.

B. Learning-Guided Semi-Sparsity (LGSS)

While the adaptive extension derives spatial weights from internal gradient computations, external priors can provide complementary information. We introduce learning-guided semi-sparsity (LGSS), which incorporates an edge probability map $E(x, y) \in [0, 1]$ into the optimization framework.

Edge map generation. The edge map is computed using the following pipeline:

1) Convert the noisy input I to grayscale (if color).
2) Apply Canny edge detection with thresholds $(T_{\text{low}}, T_{\text{high}}) = (40, 120)$.
3) Smooth the binary edge map with a Gaussian kernel G_σ with $\sigma = 3$ and kernel size 11×11.
4) Normalize to $[0, 1]$.

This produces a soft edge probability map where $E(x, y) \approx 1$ indicates high edge probability.

Weight map construction. We define the weight map as:

$$W(x, y) = 1 - E(x, y) \tag{18}$$

such that $W(x, y) \approx 0$ at edges and $W(x, y) \approx 1$ in smooth regions.

Integration into HQS. The weight map modulates the sparsity threshold:

$$\text{threshold}(x, y) = \frac{\beta \cdot W(x, y)}{\lambda} \tag{19}$$

The g-subproblem becomes:

$$g(x, y) = \begin{cases} \nabla^2 S(x, y) & \text{if } \|\nabla^2 S(x, y)\|^2 \geq \frac{\beta \cdot W(x,y)}{\lambda} \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

Rationale: At edge locations ($E \approx 1$, $W \approx 0$), the threshold is near zero, allowing second-order gradients to persist. In smooth regions ($E \approx 0$, $W \approx 1$), the threshold is β/λ, enforcing sparsity as in the original formulation.

Advantages and limitations. The use of LGSS eliminates the dependency between the prior and the internal optimization state and allows the use of more sophisticated edge detection methods or learned representations. Nevertheless, the quality of the edge detection map affects the smoothing process significantly. Noisy edges can cause degradation of the smoothing process, while missing edges can cause the smoothing process to over-smooth the images. Canny edge detection on the noisy images causes detection errors, especially in low SNR images. Although the current approach uses classical edge detection methods, the term learning-guided refers to the overall paradigm of using the learned priors in the optimization process.

C. Multi-Scale Semi-Sparsity

Multi-scale approaches have proven effective in various optimization problems, particularly for avoiding local minima and accelerating convergence. We explore a coarse-to-fine strategy:

1) **Coarse level:** Downsample the input I by a factor of 2 to obtain I_{coarse}.
2) Run the original semi-sparsity solver on I_{coarse} to obtain S_{coarse}.
3) Upsample S_{coarse} to the original resolution via bilinear interpolation, obtaining S_{init}.
4) **Fine level:** Run the semi-sparsity solver on I with initialization $S = S_{\text{init}}$ instead of $S = I$.

Failure analysis. Contrary to expectations, the multi-scale approach significantly degrades performance. On the Strip image, PSNR drops from 46.99 dB (original) to approximately 35–40 dB (multi-scale). We identify the following fundamental issues:

- **Global optimization structure:** The FFT-based solver of the HQS algorithm uses periodic boundary conditions and optimizes the entire image globally. Downsampling

destroys the relationship between the Fourier coefficients at different scales.

- **Initialization bias:** The continuation strategy of the HQS algorithm aims at initializing with a nearly quadratic problem (small λ) and then gradually increasing the level of sparsity. Initializing with a smoothed coarse estimate of the image biases the optimization towards an intermediate level of sparsity.
- **Aliasing artifacts:** Downsampling without proper anti-aliasing filters introduces high-frequency artifacts during optimization.

This negative result highlights a critical insight: not all algorithmic extensions are compatible with the underlying optimization framework. Multi-scale strategies are highly effective in local filtering methods and certain variational formulations but fundamentally conflict with the global FFT-based structure of semi-sparsity methods.

V. Experimental Setup

A. Dataset

We evaluate our methods on four standard grayscale test images:

- **Lena** (512×512): A portrait image with smooth facial regions and textured hat/feathers. Widely used in image processing benchmarks.
- **Cameraman** (256×256): A natural scene with a human subject and complex background. Contains both smooth regions and fine details.
- **Barbara** (512×512): A texture-rich image with clothing patterns, facial features, and striped backgrounds. Notoriously challenging for smoothing methods due to high-frequency texture.
- **Strip** (synthetic, dimensions vary): A synthetic test pattern with controlled geometric structures. Provides a controlled setting for evaluating edge preservation.

B. Noise Model

We simulate additive white Gaussian noise:

$$I = I_0 + n, \quad n \sim \mathcal{N}(0, \sigma^2) \tag{21}$$

where the noise standard deviation σ is chosen to produce realistic degradation while maintaining sufficient signal-to-noise ratio for meaningful denoising. For all experiments, we use $\sigma = 20$ (on an 8-bit scale, i.e., pixel values in $[0, 255]$), corresponding to moderate noise.

C. Evaluation Metric

We measure reconstruction quality using Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \log_{10} \frac{\text{MAX}^2}{\text{MSE}} \tag{22}$$

where $\text{MAX} = 1$ (for images normalized to $[0, 1]$) and

$$\text{MSE} = \frac{1}{NM} \sum_{x,y} (S(x, y) - I_0(x, y))^2 \tag{23}$$

PSNR is computed on cropped regions (excluding a 12-pixel border) to avoid boundary artifacts from periodic convolution.

TABLE I
PSNR (dB) Comparison of Semi-Sparsity Extensions

Image	Original	Adaptive	LGSS
Strip	46.9908	**47.0109**	44.7575
Cameraman	26.0484	**27.6249**	27.4356
Barbara	26.2961	26.1542	26.1709
Lena	24.8707	**29.3200**	28.8836

D. Implementation Details

All methods are implemented in MATLAB. The FFT-based HQS solver uses MATLAB's built-in `fft2`, `ifft2`, and `psf2otf` functions for efficiency. Edge maps for LGSS are generated using OpenCV's Canny implementation via a Python script.

Parameters:

- Original semi-sparsity: $\alpha = 0.1$, $\beta = 0.02$, $\lambda_0 = 10\beta$, $\kappa = 1.2$, $\tau = 0.95$, $\lambda_{\max} = 10^8$, $\text{iter}_{\max} = 500$.
- Adaptive: Same as original, with additional $\gamma = 10$, $\eta = 2$.
- LGSS: Same as original, with edge map threshold parameters $(T_{\text{low}}, T_{\text{high}}) = (40, 120)$ and Gaussian smoothing $\sigma = 3$, kernel size 11×11.

Parameters were selected based on preliminary experiments and kept fixed across all test images to evaluate generalization performance.

All implementations were validated by reproducing the baseline semi-sparsity results and ensuring consistent convergence behavior across all test cases.

VI. Results and Discussion

A. Quantitative Results

Table I presents PSNR values for all methods on the four test images. All values are computed on the cropped region to exclude boundary effects.

B. Overall Observations

Across all experiments, a common pattern emerges, whereby a significantly improved performance is obtained when using a spatially adaptive regularization approach, compared to its uniform counterparts. The adaptive semi-sparse approach demonstrates superior performance, achieving the highest PSNR value among all methods in three out of four test cases, and recording the largest improvement in terms of natural images with heterogeneous structures. Although competitive results are obtained by the LGSS approach, they do not surpass those obtained by the adaptive approach, implying that edge information obtained from external sources may not necessarily contain relevant information when gradient-based cues are employed. On the other hand, the multi-scale approach, although successful in other image processing scenarios, fails to perform adequately, largely due to its incompatibility with the global optimization scheme employed in the semi-sparsity framework.

The results obtained indicate that the main drawback of the original model lies in its inability to incorporate a spatially

Fig. 1. Comparison of denoising results on Lena image. From left to right: Noisy image, Original semi-sparsity, Adaptive method, and LGSS. Adaptive method preserves fine structures more effectively while suppressing noise.

adaptive regularization scheme, and not in its ability to incorporate structural priors effectively. Therefore, a more effective extension to the original model would be one that aims to improve its local regularization ability, rather than its global optimization scheme.

Notably, the adaptive method achieves the largest improvement of +4.45 dB on the Lena image, representing the most significant gain across all experiments and demonstrating its effectiveness on images with mixed smooth and textured regions.

Although comparisons are restricted to semi-sparse variants, previous studies have shown that L_0 gradient minimization and TV-based methods can attain a PSNR value within the range 25-28 dB for standard images [4], [9].

Overall, the adaptive approach has emerged as a superior approach, thus implying that gradient-based internal derivation of spatial modulation is more reliable than externally derived edge information for use with the semi-sparse framework.

As illustrated in Fig. 1, the adaptive method better preserves structural details while suppressing noise compared to both the original and LGSS approaches.

C. Analysis by Method

Original semi-sparsity. The baseline method achieves strong performance on the synthetic Strip image (46.99 dB), demonstrating excellent preservation of geometric structures. However, performance on natural images is more modest (24.87–26.30 dB), indicating limitations in handling complex texture and noise patterns.

Adaptive extension. The adaptive method provides consistent improvements on three out of four images:

- **Lena**: +4.45 dB improvement, the largest gain among all test cases. The adaptive threshold successfully preserves facial features and hat texture while aggressively smoothing background regions.
- **Cameraman**: +1.58 dB improvement. The method handles the complex background more effectively than the original formulation.
- **Strip**: +0.02 dB marginal improvement. The synthetic nature of this image means edges are already well-defined, leaving little room for adaptive gains.
- **Barbara**: -0.14 dB slight degradation. The texture-rich content causes the adaptive mechanism to become overly conservative, failing to smooth texture patterns that should be classified as noise.

Overall, the adaptive extension demonstrates the best generalization across diverse image types, with particularly strong performance on images with clear foreground-background separation (Lena, Cameraman).

LGSS. The learning-guided approach achieves strong results on Lena (28.88 dB), representing a +4.01 dB improvement over the baseline, though slightly lower than the adaptive method's 29.32 dB. On Cameraman (27.44 dB), it performs competitively but falls slightly short of adaptive. However, on Strip (44.76 dB), LGSS exhibits a significant -2.24 dB degradation.

This inconsistency stems from edge map quality:

- **Lena and Cameraman**: The Canny detector identifies most salient edges reasonably well, providing useful guidance. However, the reliance on gradient-based edge detection introduces some redundancy with the adaptive method's internal gradient weighting, explaining why LGSS does not surpass adaptive despite the additional prior information.
- **Strip**: The synthetic geometric patterns produce ambiguous Canny responses, leading to misclassified regions. The Gaussian smoothing of the edge map further blurs the distinction between edges and smooth areas.
- **Barbara**: The dense texture patterns trigger numerous false edge detections, confusing the LGSS framework.

LGSS demonstrates the potential of incorporating external priors into the semi-sparsity framework; however, the current implementation based on Canny edge detection does not introduce sufficiently complementary information to consistently outperform adaptive gradient-based modulation. Future work could explore learned edge detectors (e.g., HED, RCF) trained on denoising-specific objectives to improve robustness.

D. Analysis by Image

Lena. Both methods (Adaptive, LGSS) show significant improvements over the original approach (greater than 4 dB). The image includes smooth areas (skin, background) and textured areas (hat, feathers), making it a good image for testing the advanced methods. The best PSNR is obtained with the adaptive method (29.32 dB), closely followed by

the LGSS method (28.88 dB). The high PSNR values (above 28 dB) for the advanced methods indicate that the image is reconstructed almost perfectly, without significant over-smoothing of semantic textures. The high performance of the adaptive method also confirms that gradient-based spatial modulation is sufficient to capture the major variations in this image, and additional edge information is not necessary.

Cameraman. The image quality of the Cameraman image is improved significantly by the Adaptive method (+1.58 dB), indicating that the complex background and the human figure require spatially adaptive strengths of the smoothing filter. The performance of the LGSS method is also satisfactory, with an improvement of +1.39 dB.

Barbara. The texture-rich image Barbara is challenging for all methods, and the PSNR values are barely above 26 dB. The fundamental limitation is the second-order sparsity methods, as texture patterns have high second-order gradients (curvature) and are, hence, indistinguishable from semantic edges. The Adaptive method and the LGSS method are unable to handle the ambiguity, as they are based only on second-order sparsity.

The slight deterioration in PSNR values for the Adaptive method (-0.14 dB) and the LGSS method (-0.12 dB) suggests that the spatial modulation mechanisms are confused by the texture patterns and tend to preserve the noise.

Strip. The synthetic image provides an informative edge case. The original method achieves near-perfect reconstruction (46.99 dB), indicating that the global threshold is well-suited for geometric structures. The Adaptive method is also close, with an improvement of +0.02 dB. However, the LGSS method is far less accurate, with a deterioration of 2.24 dB, indicating that Canny edge detection is not reliable for synthetic patterns.

E. Failure Analysis: Multi-Scale Approach

As detailed in Section IV, the multi-scale extension fails catastrophically. On the Strip image, the coarse-to-fine approach yields PSNR values in the range of 35–40 dB, compared to 46.99 dB for the original method. Visual inspection reveals severe smoothing artifacts and loss of geometric fidelity.

This failure underscores a critical principle: *algorithmic extensions must be compatible with the underlying optimization framework*. The FFT-based HQS solver inherently assumes a single-scale, globally coherent representation. Multi-scale decompositions that work well in local filtering (e.g., bilateral filtering, guided filtering) or wavelet-based methods fundamentally conflict with the Fourier domain structure of semi-sparsity.

F. Qualitative Observations

Beyond PSNR, qualitative observations reveal important behavioral differences:

- **Edge sharpness**: Adaptive and LGSS produce visibly sharper edges compared to the original method, particularly on Lena's facial features and the Cameraman's silhouette.
- **Texture preservation**: All methods struggle to distinguish semantic texture (e.g., Barbara's clothing) from noise, often over-smoothing fine patterns.
- **Staircase artifacts**: Unlike TV methods, semi-sparsity does not exhibit pronounced staircase artifacts, maintaining smooth linear gradients in appropriate regions.
- **Boundary effects**: The periodic boundary conditions of FFT occasionally introduce wrap-around artifacts near image borders, mitigated by boundary cropping during PSNR evaluation.

VII. Applications

The proposed extensions to semi-sparsity have broad applicability across image processing and computer vision tasks:

Image denoising. The most prominent application demonstrated in the paper. Both adaptive and LGSS approaches are capable of efficient noise removal while maintaining fine image features, which is particularly important in medical imaging, satellite imaging, and low-light photography.

Texture removal and semantic abstraction. The semi-sparse approach has an inherent ability to reduce textures while maintaining semantic edges, which is particularly suitable for non-photorealistic rendering, image stylization, and content-aware image abstraction. The adaptive approach extends the semi-sparse approach to improve the removal of textures.

Edge detection preprocessing. Many edge detection algorithms, including Canny, Sobel, etc., are highly sensitive to noise in the image. Applying the semi-sparse approach as a preprocessing technique can improve the accuracy of the edge detection algorithm. Moreover, the LGSS approach provides a feedback mechanism, allowing the edge detection algorithm to improve the smoothing, which in turn can improve the accuracy of the edge detection algorithm.

Image compression. The piecewise-linear approximation obtained from the semi-sparse approach can be an efficient representation of the image, which can be used in lossy image compression.

Depth map filtering. Depth sensors are noisy, but the depth maps obtained from these sensors are likely to have sharp discontinuities near object boundaries. The semi-sparse approach has an inherent ability to maintain edges, which can be particularly suitable for depth map smoothing.

Medical image processing. In medical imaging, including CT, MRI, ultrasound, etc., the image features need to be preserved while removing noise in the image, which is particularly important in the accurate interpretation of the image.

VIII. Conclusion

This work presented a systematic investigation of three extensions to the semi-sparsity framework for edge-preserving image smoothing: adaptive gradient-order selection, learning-guided semi-sparsity (LGSS), and multi-scale decomposition. The results lead to several key insights.

Spatial adaptivity is the dominant factor for performance improvement. The adaptive extension consistently achieves the best results across diverse image types, with

substantial gains on natural images (Lena: +4.45 dB, Cameraman: +1.58 dB). This demonstrates that locally modulating the regularization strength is critical for handling heterogeneous image structures. Importantly, this improvement is achieved with minimal modification to the original optimization framework. However, the adaptive method exhibits clear limitations in texture-rich regions such as Barbara, where second-order sparsity fails to distinguish between noise and structured texture, resulting in performance degradation and revealing a fundamental limitation of the formulation.

External priors must provide complementary information. The LGSS approach introduces a hybrid framework by incorporating edge-based priors. While it achieves competitive performance, it does not surpass the adaptive method, indicating that gradient-based edge detection provides information largely redundant with the model's internal gradient structure. This highlights the need for more informative priors, such as semantic or learned representations, to achieve further gains.

Algorithmic extensions must respect optimization structure. The failure of the multi-scale approach reveals that modifications incompatible with the global Fourier-based optimization can significantly degrade performance. This underscores the importance of aligning algorithmic innovations with the mathematical structure of the underlying solver.

Texture modeling remains an open challenge. All methods exhibit limited performance on texture-rich images such as Barbara, indicating that second-order sparsity alone is insufficient to distinguish semantic texture from noise. Addressing this limitation requires integration of higher-level priors or texture-aware models.

In conclusion, this study demonstrates that spatially adaptive regularization is a highly effective strategy for improving semi-sparsity-based smoothing, while also identifying key limitations and directions for future research. Future work should explore integration with learned priors that provide complementary information beyond gradient-based cues, as well as principled approaches to automatic parameter selection.

IX. Future Work

Several promising directions emerge from this investigation:

Learned edge priors. Replace handcrafted edge detectors such as the Canny edge detector with learned edge detectors such as HED, RCF, or BDCN, or learned edge uncertainty maps. Train edge networks on noisy images to improve performance for low SNR scenarios.

Joint optimization of priors and smoothing. Replace the separate edge estimation step with a joint optimization problem that iteratively estimates edges and smooths the image. This may allow for a refinement process, where improved smoothing results in improved edge estimation, and vice versa.

Texture-aware regularization. Add texture features such as local binary patterns or Gabor filter responses to distinguish semantic textures from noise. A three-class classification problem (smooth, edge, and texture) may allow for texture-preserving smoothing.

Deep learning integration. Train a CNN to output spatially varying parameter maps for $\alpha(x, y)$, $\beta(x, y)$, and $\lambda(x, y)$. This may allow for automatic parameter tuning and task-specific optimization (different parameter maps for different applications, e.g., denoising and abstraction).

Alternative optimization frameworks. Consider alternative optimization methods such as primal-dual methods or ADMM, potentially allowing for a multiresolution extension or non-periodic boundary conditions. Investigate sparse solvers that do not rely on the FFT to eliminate periodic boundary artifacts.

Higher-order extensions. Extend the framework to third-order or higher-order derivatives to promote piecewise polynomial approximations. This may improve the representation power for smooth gradients and curved surfaces.

Color image processing. Investigate principled approaches for color image smoothing, including correlations between color channels and color spaces such as LAB or YUV. The current grayscale approach may not generalize optimally to color images.

Computational efficiency. Investigate GPU acceleration for the FFT-based solver to allow for real-time smoothing of high-resolution images or video streams. Consider approximations such as down-sampling gradients to reduce computational cost without sacrificing smoothing quality.

Theoretical analysis. Provide theoretical guarantees for convergence of the HQS algorithm for spatially varying parameters. Investigate the optimization landscape to ensure that local minima are avoided.

Application-specific tuning. Extend the smoothing framework to application-specific areas such as medical imaging, remote sensing, or computational photography, including application-specific priors and evaluation criteria.

Acknowledgment

The authors would like to thank Dr. Jignesh Patel of the Indian Institute of Information Technology Vadodara for providing computational resources and support for this research. We acknowledge the original authors of the semi-sparsity method for their foundational work, which inspired this further study.

References

[1] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 1998, pp. 839–846.

[2] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 35, no. 6, pp. 1397–1409, 2013.

[3] P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 12, no. 7, pp. 629–639, 1990.

[4] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D: Nonlinear Phenomena*, vol. 60, no. 1-4, pp. 259–268, 1992.

[5] T. Goldstein and S. Osher, "The split Bregman method for L1-regularized problems," *SIAM Journal on Imaging Sciences*, vol. 2, no. 2, pp. 323–343, 2009.

[6] A. Chambolle and T. Pock, "A first-order primal-dual algorithm for convex problems with applications to imaging," *Journal of Mathematical Imaging and Vision*, vol. 40, no. 1, pp. 120–145, 2011.

[7] K. Bredies, K. Kunisch, and T. Pock, "Total generalized variation," *SIAM Journal on Imaging Sciences*, vol. 3, no. 3, pp. 492–526, 2010.

[8] J. Cai, S. Osher, and Z. Shen, "Split Bregman methods and frame based image restoration," *Multiscale Modeling & Simulation*, vol. 8, no. 2, pp. 337–369, 2009.

[9] L. Xu, C. Lu, Y. Xu, and J. Jia, "Image smoothing via L_0 gradient minimization," *ACM Trans. Graphics*, vol. 30, no. 6, pp. 174:1–174:12, 2011.

[10] Q. Xin, Z. Liu, and X. Wang, "Semi-Sparsity for Smoothing Filters," *arXiv preprint arXiv:2107.00627*, 2021.

[11] M. Lysaker, A. Lundervold, and X. Tai, "Noise removal using fourth-order partial differential equation with applications to medical magnetic resonance images in space and time," *IEEE Trans. Image Processing*, vol. 12, no. 12, pp. 1579–1590, 2003.

[12] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Processing*, vol. 26, no. 7, pp. 3142–3155, 2017.